ocean currents. Temperature gradients are usually high over the western margins of continents. This is because of cold ocean currents.

Statement 2: CORRECT—Low temperature gradients are observed over tropics (sun is almost overhead the entire year) and high temperature gradients over middle and higher latitudes (sun's apparent path varies significantly from season to season).

65. **Statement 1: INCORRECT**—The isotherms are irregular over the northern hemisphere due to an enhanced land-sea contrast. Because of predominance of land over water in the north, the northern hemisphere is warmer. The thermal equator (ITCZ) lies generally to the north of geographical equator.

Statement 2: CORRECT—While passing through an area with warm ocean currents, the isotherms show a poleward shift. (North Atlantic Drift and Gulf Stream combined with westerlies in Northern Atlantic; Kurushio Current and North Pacific current combined with westerlies in Northern Pacific).

66. Latent Heat: Latent heat of condensation is the driving force behind all tropical cyclones.

It is the heat released or absorbed during phase change.

Latent heat, characteristic amount of energy absorbed or released by a substance during a change in its physical state that occurs without changing its temperature.

The latent heat associated with melting a solid or freezing a liquid is called the heat of fusion; that associated with vapourising a liquid or a solid or condensing a vapour is called the heat of vapourisation.

The latent heat is normally expressed as the amount of heat (in units of joules or calories) per mole or unit mass of the substance undergoing a change of state.

68.